```python
#sum of 1st n natural numbers
sum = 0
n=int(input("Enter no of elements : "))
for i in range(1,n+1):
    sum+=i
print(sum)
```

```
Enter no of elements : 10
55
```

```python
#count number of even and odd numbers in the given series
p=int(input("enter p :"))
q=int(input("enter q :"))
even = 0
odd = 0
for i in range(p,q+1):
    if(i%2 == 0):
        even+=1
    else:
        odd+=1

print(f"Total Even numbers : {even}")
print(f"Total Odd numbers : {odd}")
```

```
enter p :51
enter q :60
Total Even numbers : 5
Total Odd numbers : 5
```

```python
#prime checker
n = int(input("Enter a number : "))
i = 2
count = 0

while (i < n):
    if (n % i == 0):
        count += 1
        break
    i += 1

if (count == 0 and n > 1):
    print("Prime")
else:
    print("Not prime")
```

```
Enter no : 4
```

```python
#fibonnaci
n = int(input("Enter the last term u want in fibonacci series : "))
a=0
b=1
```

```
while(a <= n):
    print(a,end=" ")
    a,b=b,a+b
```

```
Enter the last term u want in fibonacci series : 13
0 1 1 2 3 5 8 13
```

```
#nested pattern loop
for i in range(1,6):
    for j in range(6-i):
        print(" ",end="")
    for k in range(2*i-1):
        print("*",end="")
    print()
```

```
     *
    ***
   *****
  *******
 *********
```

```
#pattern 1
for i in range(1,6):
    for j in range (6-i):
        print("*",end=" ")
    print()
```

```
* * * * *
* * * *
* * *
* *
*
```

```
#pattern 2
for i in range(1,6):
    for j in range(1,i+1):
        print(j,end="")
    print()
```

```
1
12
123
1234
12345
```

```
#pattern 3
n1 = 3
for i in range(n1):
    for j in range(i+1):
        print("* ",end="")
    print()
```

```python
for i in range(n1):
    for j in range(i,n1-1):
        print("* ",end="")
    print()
```

```
*
* *
* * *
* *
*
```

```python
#fizzbuzz
for i in range(1,51):
    if(i%3==0 and i%5!=0):
        print("Fizz")
    elif(i%3!=0 and i%5==0):
        print("Buzz")
    elif(i%3==0 and i%5==0):
        print("FizzBuzz")
    else:
        print(i)
```

```
1
2
Fizz
4
Buzz
Fizz
7
8
Fizz
Buzz
11
Fizz
13
14
FizzBuzz
16
17
Fizz
19
Buzz
Fizz
22
23
Fizz
Buzz
26
```

```
Fizz
28
29
FizzBuzz
31
32
Fizz
34
Buzz
Fizz
37
38
Fizz
Buzz
41
Fizz
43
44
FizzBuzz
46
47
Fizz
49
Buzz
```

```
#string operations
s = input("enter string sentence: ")
print(s.upper())
print(s.lower())
print(s.capitalize())
s=s.split()
print(s)
s='_@_'.join(s)
print(s)
```

```
enter string sentence: Hello world i am python not language snake
HELLO WORLD I AM PYTHON NOT LANGUAGE SNAKE
hello world i am python not language snake
Hello world i am python not language snake
['Hello', 'world', 'i', 'am', 'python', 'not', 'language', 'snake']
Hello_@_world_@_i_@_am_@_python_@_not_@_language_@_snake
```

```
#continue
for i in range(10):
    if(i%2==0):
        continue
    print(i)
```

```
1
3
```

```
5
7
9

#pass and else
for i in range(11):
    if(i==3):
        pass
    print(i)
else:
    print("Loop complete")

0
1
2
3
4
5
6
7
8
9
10
Loop complete
```